This documentation outlines the process of extracting data from a series of `.sum` files and saving the extracted data in a CSV format in a well-documented manner. A framework consisting of three Python scripts has been developed to achieve this task. The step-by-step workflow is described here to ensure that others can use this framework for future applications.

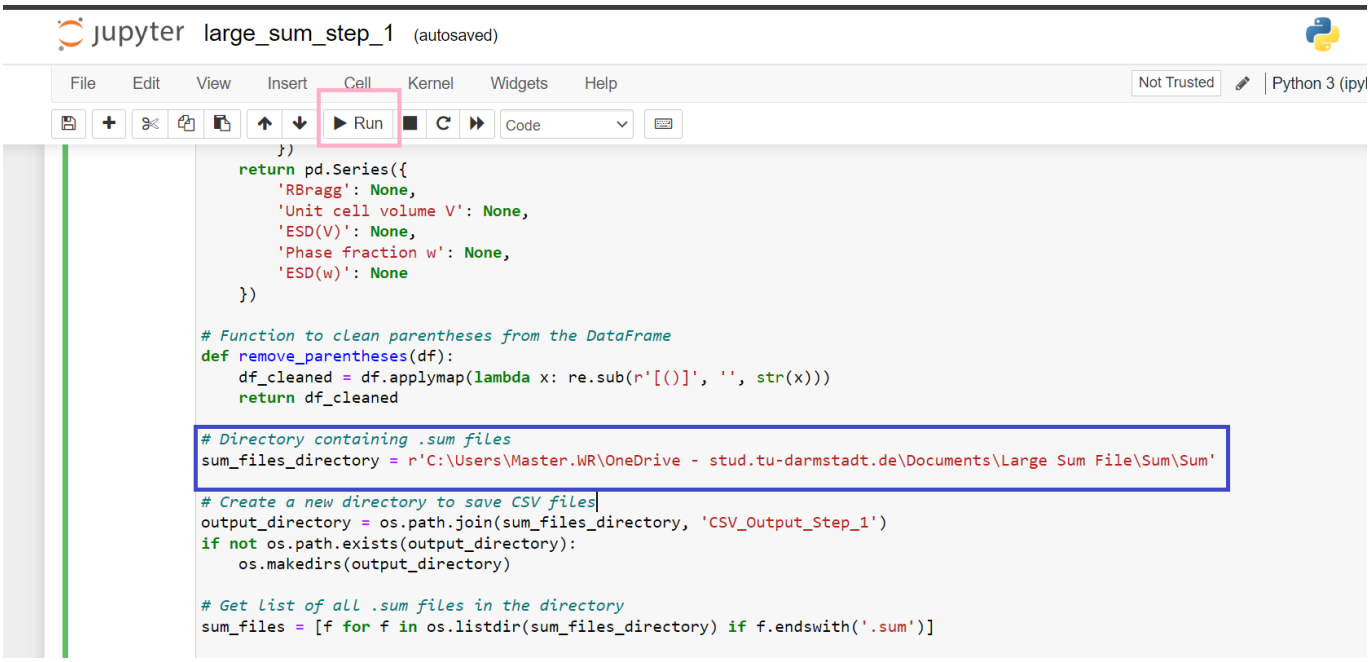
Download the Scripts: Begin by downloading the following three Python scripts: `large_sum_step_1.ipynb`, `large_sum_step_2.ipynb`, and `large_sum_step_3.ipynb`. Save these scripts in the `User` folder on your computer. Additionally, download or copy the `.sum` files into the `User` folder. Alternatively, you may store the `.sum` files in a subfolder within the `User` folder.

Open Jupyter Notebook: If Jupyter Notebook is already installed on your system, open it by searching for "Jupyter Notebook" in the Start menu. Once open, navigate to and open the file named `large_sum_step_1.ipynb`. This file will open in a new browser tab.

Provide the Path to Your `.sum` File

In the script, you will need to specify the path to your `.sum` file. Locate the line in the script where the file path is required. In the accompanying image, this line is highlighted within a blue rectangular box. Provide the correct path to your `.sum` file, ensuring that the path is enclosed within single quotation marks (`''`).

Once you have entered the path, click on the "Run" option, which is indicated by a pink rectangle in the image.



Running the Second Script

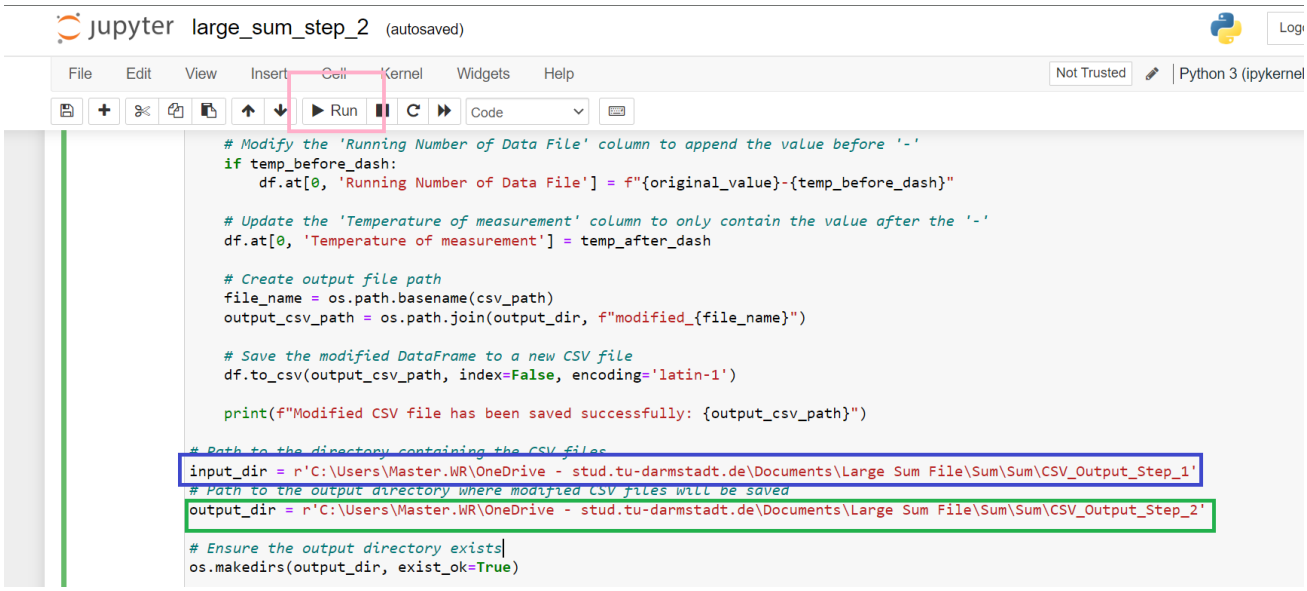
After running the first script (`large_sum_step_1.ipynb`), a set of CSV files will be generated and saved in the same directory. You can organize these files by moving them into a separate folder

within the same directory.

Next, open the second script named `large_sum_step_2.ipynb`. In this script, you will need to modify two specific fields as marked in the accompanying image:

- **Blue Rectangular Box**: In this section, provide the path to the CSV files that were generated after running the first script. Ensure that the path is enclosed within single quotation marks (`''`).
- **Green Rectangular Box**: Here, specify the path where you would like to save the new set of CSV files that will be generated after running this script. Before doing so, create a new folder within the same directory to store these CSV files. Provide the full path to this new folder, again enclosed in single quotation marks (`''`).

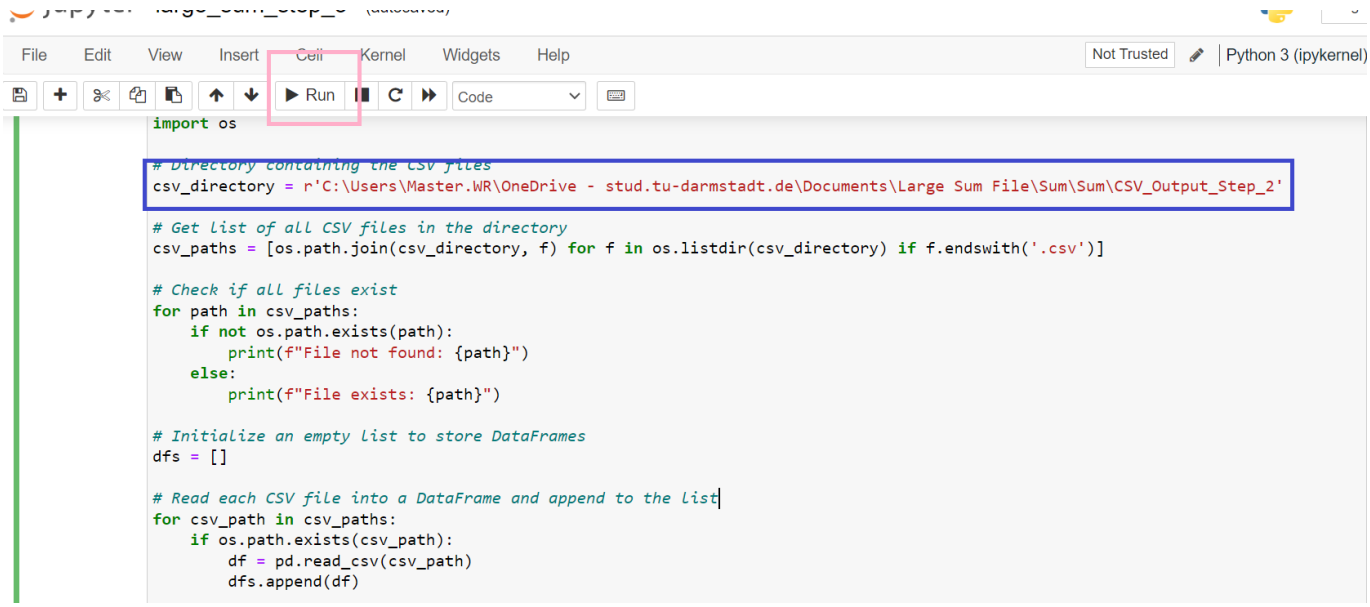
  And then click "Run" option, which is indicated by a pink rectangle in the image.



Running the Final Script

Now, open the last script named `large_sum_step_3.ipynb`. In this script, you only need to provide the path to the folder where you saved all the CSV files generated after running the second script. Ensure the path is enclosed within single quotation marks (`''`).

After entering the correct path, run the script. Once the script has finished executing, navigate to the specified folder. You will find a new CSV file named `large_sum_combined_data.csv`, which is your final desired output.


File
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Python 3 (ipykernel)
Run
Code
import os
Directory containing the CSV files
csv_directory = r'C:\Users\Master.WR\OneDrive - stud.tu-darmstadt.de\Documents\Large Sum File\Sum\Sum\CSV_Output_Step_2'
Get list of all CSV files in the directory
csv_paths = [os.path.join(csv_directory, f) for f in os.listdir(csv_directory) if f.endswith('.csv')]
Check if all files exist
for path in csv_paths:
if not os.path.exists(path):
print(f"File not found: {path}")
else:
print(f"File exists: {path}")
Initialize an empty list to store DataFrames
dfs = []
Read each CSV file into a DataFrame and append to the list
for csv_path in csv_paths:
if os.path.exists(csv_path):
df = pd.read_csv(csv_path)
dfs.append(df)